

THE PRODUCTIVE AND PSYCHOSOCIAL IMPACTS OF A CASH PLUS HEALTH INSURANCE INTERVENTION: EVIDENCE FROM GHANA

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Abstract

We examine the effect of LEAP 1000, an integrated cash transfer and health insurance intervention on livestock, productive investments and behavioural outcomes in Ghana. Using a two-wave panel data set and leveraging a proxy means test targeting, household fixed-effects regressions show that the intervention increased livestock ownership and commercialization. LEAP 1000 also increased savings and participation in wage employment. Delving into what could be driving these findings, we find that investments in livestock and other productive investments are driven by behavioral and psychosocial traits such as orientations towards the future or present preferences, expectations about the future, happiness, and agency. Local average treatment effects show that receiving cash plus insurance dominated the effect of the intervention.

Keywords: Cash Transfers; Health Insurance, Household Investments; Future Orientations; Ghana

JEL Codes: I13, I15, I18, J01

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1 Introduction

The growing empirical literature supports the premise that social assistance programs, especially cash transfers, have lifted many households out of poverty and increased shared prosperity (Alderman & Yemtsov, 2013; Banerjee et al., 2015; Daidone, Davis, Handa, & Winters, 2019; Fisher et al., 2017; Handa et al., 2018; Skoufias & Di Maro, 2008). Despite these advances, it is also increasingly accepted that single social assistance interventions are not a panacea and that concerted integrated interventions will be required for better livelihood and welfare outcomes (Balboni, Bandiera, Burgess, Ghatak, & Heil, 2022; Bossuroy et al., 2022; Daidone et al., 2019). As such, there is an increase in integrated social assistance programs that target multiple vulnerabilities (Bossuroy et al., 2022). For example, in Ethiopia, the Productive Safety Net Program (PSNP) and community health insurance are increasingly combined (Shigute et al., 2017; Shigute, Strupat, Burchi, Alemu, & Bedi, 2020) and in Rwanda, extremely poor households benefit from both health insurance premium waivers and are targeted for cash transfers and other interventions under the country's flagship Vision2020 Umerenge Program (VUP). Despite the increasing focus on these programs and their promise, there is still a dearth of evidence on the progressive, additive impacts of these integrated programs on welfare and graduation from poverty. This paper is about one such program, the Ghana Livelihood Empowerment against Poverty (LEAP) 1000 program.

In this study, we investigate the productive impact of this joint unconditional cash transfer and health insurance program on productive livelihoods. The LEAP 1000 is an extension of the flagship Livelihood Empowerment Against Poverty (LEAP) cash transfer program that targets pregnant women and families with children aged 0-1 in Ghana. This extension, in addition to providing households (with children under 2 years of age or pregnant women) with cash transfers, also provides health insurance support by paying for their enrollment in the Ghana National Health Insurance Scheme. We examine the association between LEAP 1000 and (1) livestock ownership; (2) livestock commercialization and (3) savings and loans, (4) investments in durable goods, (5) starting a business and (6) engaging in wage employment. In making these productive investment decisions, we hypothesize that psychological factors such as future orientation, expectation, and socio-emotional skills such as patience, happiness, and agency mediate the relationship between LEAP 1000 and household productive investments. We rely

on a two-wave panel data set from Ghana that included a time preference survey experiment. Empirically, we employ a household fixed-effects model to assess intent to treat and weighted least squares fixed effects to estimate the late average treatment effects.

We report several key results. First, we show that LEAP 1000 increases the ownership of livestock by about 7.9 percentage points (15%). Some of the livestock that respond to LEAP 1000 are small ruminants such as goats (8.2 percentage points) and poultry (9.1 percentage points). Second, we also observe that LEAP 1000 increases livestock commercialization (purchase and sale), the probability of saving, as well as wage employment. Third, LEAP 1000 increases present orientation and impatience, happiness, expectation about the future, and agency. In contrast, LEAP 1000 reduces future orientation, making individuals focus on the present, a finding we argue is driven by the existing inflation in Ghana at the time, which made individuals averse to future orientation. The above insights are based on the intention-to-treat effects (ITT), where we focus just on the impact of the cash transfer component of LEAP 1000. We also estimate local average treatment effect regressions where we considered both the combined cash and insurance treatments. We find very similar results, which are consistent with the ITT insights. In the interest of testing behavioral and psychosocial traits as potential pathways explaining the positive impact of LEAP 1000 on livestock and non-farm investments, we interact each pathway with the treatments. In general, we find the important role that these behavioral and psychological traits play in mediating productive investments and employment.

Our analysis offers three contributions to different strands in the empirical literature on cash transfers and welfare improvements in developing countries. The first contribution is to highlight the crucial role of cash transfers and health insurance in productive investments and employment ([Aggarwal et al., 2024](#); [Gertler, Martinez, & Rubio-codina, 2012](#); [Stoeffler, Mills, & Premand, 2020](#)). We establish the link and transition from social protection as a medium of protection to a more productive role that involves investments in productive assets such as livestock and savings ([Devereux & Sabates-wheeler, 2004](#); [Schuering & Loewe, 2023](#)). We also touch on the transformational role of social protection given that we consider women and children who are usually excluded from many development programs ([Devereux & Sabates-wheeler, 2004](#)). Our measurement of productive investments makes this even more interesting as we focus on livestock which is considered as rural wealth in many rural settings ([Tab-Ojong, Heckeleei, & Rasch, 2023](#)). Livestock investments are a common store of value, assets and capital and enormously contribute to building resilience in households during times of

income shocks ([Acosta, Nicolli, & Karfakis, 2021](#); [Lange & Reimers, 2021](#)). Livestock also denote cultural ties and have cultural value and are used to smooth consumption in times of shocks. Moreover, they have been increasingly associated with future orientation ([Tabe-Ojong, Heckelei, & Rasch, 2023](#)).

Second, and related to the first, we look at one of the very contested but often wrong perceptions about cash transfers, that they promote dependency and reduce productive work ([Handa et al., 2018](#); [Kalebe-Nyamongo & Marquette, 2014](#)). Previous analyses on this relationship have looked at agricultural assets as part of household investments ([Boone, Covarrubias, Davis, & Winters, 2013](#); [Covarrubias, Davis, & Winters, 2012](#)) with little focus on livestock. Some exceptions are [Asfaw, Davis, Dewbre, Handa, and Winters \(2014\)](#), [Daidone et al. \(2019\)](#) and [Shigute et al. \(2020\)](#). Using longitudinal data from Kenya and applying a difference in differences coupled with a matching estimator, [Asfaw et al. \(2014\)](#) found positive and significant effects of cash transfer on the probability of livestock ownership, the total number of livestock units, and the formation of non-farm enterprises. Similarly, [Daidone et al. \(2019\)](#) found positive effects on the probability of owning livestock in Malawi, Zambia, and Zimbabwe and small ruminant livestock in Ethiopia and Ghana. Our study is different from these two in two dimensions. First, in [Asfaw et al. \(2014\)](#), targeting was not systematic and, therefore, the baseline differences between the households receiving and those not receiving the cash transfer were large and systematic in their sample. Though they use matching to account for these differences, their causal estimates cannot account for other unobserved heterogeneity. Second, our study is different in the way that it looks at a combination of interventions (health insurance and cash transfers), while other studies have looked at only cash transfers. A closely related study is [Shigute et al. \(2020\)](#) that studies the joint effects of cash transfers and health insurance in Ethiopia. They found that neither CBHI nor health insurance alone increased the value of livestock, but when combined, the value of livestock holding increased by 4 percent. However, in their study, though they have panel data, they are unable to control endogeneity emanating from self-selection into both health insurance and participation in the cash transfer program.

Adding on the above contributions, our study explores the relationship between cash transfers and psychological well-being, which is relatively understudied. Specifically, we look at the effect of combined cash transfers and health insurance on behavioural and psychological outcomes elicited through a time preference survey experiment. Beyond economic and

productive impacts, cash transfers could also have a large impact on psychological well-being (Correa, Daidone, Davis, & Sitko, 2023; Handa, Martorano, Halpern, Pettifor, & Thirumurthy, 2016; Handa, Seidenfeld, & Tembo, 2020; Haushofer & Fehr, 2014; Ibrahim et al., 2023; Mani, Mullainathan, Shafir, & Zhao, 2013; Molotsky & Handa, 2021). These aspects have only been explicitly considered by a few studies in the literature (Al Izzati, Suryadarma, & Suryahadi, 2023; Handa et al., 2016, 2020; Ibrahim et al., 2023; Molotsky & Handa, 2021; Suarez & Cameron, 2020), but the context matters, as we highlight in our results. We examine the impacts of cash transfers on several behavioral and psychological outcomes. In addition, we also examine whether the present or future orientation could explain the relationship between program participation and investments leading to productive investments and employment. Our insights here contribute to the growing literature on the psychology of poverty that considers the potential of using psychosocial measures to unlock trajectories out of poverty (Haushofer & Fehr, 2014; Ibrahim et al., 2023; Molotsky & Handa, 2021). Furthermore, the relationship between cash transfers and psychosocial outcomes such as stress and social support is likely to mediate the effect on investment decisions by households (Haushofer & Fehr, 2014; Zheng, Carter, Jensen, & Krovetz, 2023). Previous studies have only considered these independent outcomes of cash transfers without expressly examining these as pathways linking cash transfers and outcomes.

The remainder of the paper is structured as follows. Section 2 is an overview of the LEAP 1000 cash transfer program and its evaluation design. Section 3 dives directly into the data, the measurement of various variables used in the analysis, and the empirical strategy. It also contains some additional analysis and some different robustness checks. The results are then presented and discussed in Section 4 with some triangulation and discussion carried out based on the empirical literature in Section 5. The paper concludes with a summary, policy implications and recommendations, limitations of the study, and potential gaps for future analysis in Section 6.

2 Ghana's LEAP 1000 Programme

The Livelihood Empowerment Against Poverty (LEAP), which started in 2008, is Ghana's flagship poverty alleviation program and is implemented under the guidance of the Ministry of Gender, Children, and Social Protection (MoGCSP) ([Tabe-Ojong, Boakye, & Muliro, 2020](#)). The program provides cash assistance to poor and vulnerable households bimonthly.

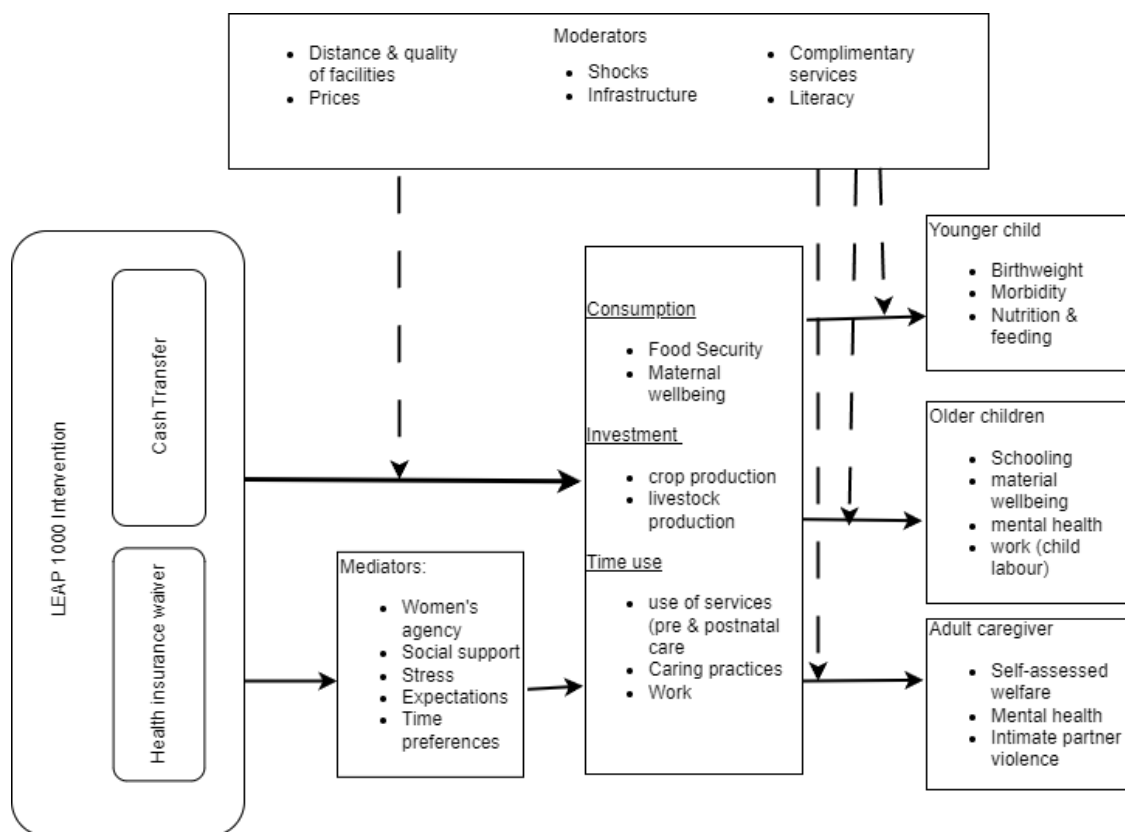
In 2015, the Government of Ghana piloted an extension of LEAP (LEAP 1000) aimed at assessing the effectiveness and impact of targeting cash transfers in the first 1000 days of a child's life, which has potential implications for in utero and early childhood health. To be eligible to participate in the intervention, a household had to have a pregnant woman or a child under 15 months of age. The evidence of these was presented by valid antenatal birth certificates or birth certificates of the children. The other condition was that households generally had to be needy or poor. To establish the second criterion, a short survey (covering household composition, educational attainment, agricultural assets, housing, durables ownership, and location) was conducted and an eligibility proxy means test was developed. Those who met the criteria were selected for the intervention ([Transfer Project, 2020](#)). In total, approximately 6124 households with eligible women were enrolled in 2015 ([Transfer Project, 2020](#)). Households with an eligible person received 64 Cedis while the amount increased by 66% to 106 Cedis per household per month.

In addition to the poverty targeting, the intervention also implemented a geographical targeting. Ten districts in northern Ghana (North East and Northern regions) were selected based on their poverty and nutrition conditions. Within each district, community poverty rankings were calculated and the poorest communities were selected if they were not already covered by the original/ mainstream LEAP program. After determining which locations would be included in the program, a short survey was implemented with the purpose of creating a proxy means test (PMT) such that poorer households would be prioritized. Indicators included in the PMT score were household composition, educational attainment, agricultural assets, housing, durables ownership, and location [LEAP 1000 Evaluation Team \(2016\)](#). Individuals below PMT were prioritized for inclusion in the program while those above the threshold were not included in the program. Approximately 6,200 households were eligible to participate.

One unique attribute of the LEAP 1000 pilot program is its 'cash +' nature, where it opens access to free health insurance coverage to beneficiaries by providing both a cash transfer and a health insurance premium waiver to participating individuals. Health insurance

coverage included access to inpatient, outpatient, dental, and maternal health services (LEAP 1000 Evaluation Team, 2018). The health insurance component of LEAP 1000 was made possible through a partnership with the Ghana National Health Insurance Scheme (NHIS). Ghana started a national health insurance scheme in 2003 and in 2014, about 66% of the population was said to be enrolled (Aboagye et al., 2023; Ayanore et al., 2019). The annual premiums are about \$15 for a family of 5 individuals (Garcia-Mandico, Reichert, & Strupat, 2021). However, unlike the convention in many low-income community insurance programs to enroll full households at once, the Ghana scheme enrolls individuals. Consequently, the scheme records partial household enrolment and increased rates of dropout (Kusi, Enemark, Hansen, & Asante, 2015). For this reason, the LEAP 1000 intervention sought to alleviate enrollment concerns for pregnant mothers and young children by providing insurance premium waivers. This intervention therefore, together with the Ethiopian health insurance scheme (Shigute et al., 2020) is one of the very few interventions that combine cash transfers and health insurance. Therefore, it creates an opportunity to assess the potential complementarities of the two social protection instruments. The theory of change of the intervention is shown in 1.

Figure 1: LEAP 1000 Theory of Change



The intervention was expected to have a direct causal affect on household consumption, investments in crops and livestock, and time use. These causal effects were expected to be mediated by improvements in psychosocial outcomes such as improving social support, reducing stress, and improving agency. Note that given that the actual intervention is cash plus health insurance, our theory of change differs slightly from the original theory of change in [LEAP 1000 Evaluation Team \(2016\)](#) in that the way in which we present the components of the intervention.

3 Data and variables

3.1 Baseline and Endline Household Surveys

The data used in the study are from the UNICEF Office of Research, Innocenti impact evaluation and were collected by the Institute of Statistical, Social and Economic Research (ISSER) of the University of Ghana together with UNICEF under the auspices of the Transfer Project before the intervention began([LEAP 1000 Evaluation Team, 2018](#)). Ethics approvals were obtained from the Institutional Review Boards of the University of North Carolina Chapel Hill, the Institutional Review Board of the Navrongo Health Research Center and the University of Ghana. The study was registered with the Registry for International Development Impact Evaluations (RIDIE) with study ID no. 55942496d53af.

For impact evaluation purposes, power calculations indicated that a sample of 2500 households was sufficient to observe a credible impact on key household welfare outcomes. Thus, for a finer comparison, households were purposively sampled along the PMT threshold so that those included in the program had more or less similar PMT scores. At baseline, conducted between July and September 2015, 2497 households were surveyed, of which 1262 were households that received cash payments (treated) and 1235 received no payments, therefore, control households. The baseline analysis shown in the Supplementary Table [A4](#) indicated that the treatment and control households were similar and balanced at baseline. In 2017, the follow-up survey successfully reached 2331 households. There was an attrition of about 6.6% of the sample (166 out of 2497 households). This attrition is not only considered low, but previous studies ([de Groot, Yablonski, & Valli, 2022](#); [Peterman, Valli, & Palermo, 2022](#)) also show that it did not affect the results analyzed. In our study, we find some differential attrition. Considering our set of control variables, households that dropped out were different in terms

of household size, primary education status of the mothers, household electricity, safe water access of the households, and whether an interpreter was used. To check whether differential attrition affected the results, we computed inverse probability attrition weights and included them in the model to account for this. Intent-to-treat results with these attrition weights suggest that the results remain more or less unchanged. These results are available to interested readers.

3.2 Description of Outcomes

3.2.1 Productive investments and employment outcomes

We assess the effect of LEAP1000 on household investments in two dimensions: (a) livestock investments and (b) participation / investments in non-farm livelihood activities. For livestock, we used four outcomes that include livestock ownership as an indicator variable, Total Livestock Units (TLU), calculated using conversion factors relevant for sub-Saharan Africa (Njuki et al., 2011), livestock purchase and livestock sales to show participation in livestock commercialization. TLU provides a more nuanced measure than other indicators, such as flock size, as it provides a measure to assess all combined livestock so that it matters less what type of livestock a household invests in (Njuki et al., 2011; Tabe-Ojong, Heckeleei, & Rasch, 2023). Livestock commercialization was not measured at baseline. The results we therefore present are endline single differences for purchase and selling of livestock. The households in the study area have a variety of livestock. To explain the different types of livestock a household might invest in, we divide livestock into large ruminants (cattle, donkeys) and small ruminants (sheep, goats). We also implement regressions for individual livestock, such as cattle, sheep, goats, and poultry, to further elaborate on the heterogeneity of the livestock ownership.

For nonfarm investments, we consider participation in non-farm livelihood activities in various dimensions. First, we assess investments in durable goods. Durable goods are a measure of household wealth accumulation and also act as savings vehicles. Using the household consumption module, we isolate non-food consumption of item purchases in the last 12 months. Second, we look at business enterprise ownership. Households in the study area are typically rural households and participate in many non-farm livelihood activities such as petty trading, home production activities, handicrafts, community service, shop keeping, small-scale manufacturing, and participation in various small and medium size enterprises. Third, we look at household involvement in financial markets through savings and borrowing. We therefore

assessed if a household has some savings and if it took a loan in the the last 12 months. Finally, we look at non-farm employment – if a household member engaged in any wage or periodically salaried employment.

3.2.2 Psychological outcomes and pathways

We hypothesize five pathways through which the effect of LEAP1000 on household investments/economic outcomes is likely to be reflected. First, we look at how the intervention could have affected future-oriented thinking, planning, and action. This could be done by reducing impatience, increasing future preferences, and changing the expectations that life will be better in a future period compared to the present. Therefore, we assessed the effect of the intervention on several future-oriented measures. First, three of our future-oriented measures come from a time preferences module similar to recent studies ([Handa, Martorano, Halpern, & Thirumurthy, 2014](#); [Handa et al., 2020](#); [Ibrahim et al., 2023](#); [Suarez & Cameron, 2020](#)). However, in contrast to these papers, we contextualize future orientations in the form of (a) impatience, (b) present orientations, and (c) future orientations.

In the elicitation of time preferences, individuals were presented with a hypothetical situation in which they had to choose between a certain amount of money now or a slightly higher amount of money at a future period. The higher amount in a future period factors into the measure of waiting (patience) such that waiting is always more financially rewarding with higher payouts than the present. The elicitation of time preferences was posed in a manner that ensures both preferences and their consistency such that, a opposed to only preferences where iterations are sequentially presented with consistently higher discount rates for the future, our elicitation mixes higher and lower discount rates in alternate sequences such that preferences are more consistent irrespective of discount rates. In this way, our elicitation measures are more conceptually stable and consistent than those in [Suarez and Cameron \(2020\)](#) and [Chuang and Schechter \(2015\)](#), although similar to related studies in the Transfer Project ([Handa et al., 2020](#); [Ibrahim et al., 2023](#); [Molotsky & Handa, 2021](#)). The time preference elicitation module is shown in the Supplementary Table A2. In these elicitations, individuals who select the present option are highlighted as the least patient, hence most impatient while the more future options a respondent selects the higher they levels of patience. From here, we created three measures: (1) the present orientation shows individuals who, no matter the discount rate indicated, the respondent always chooses to have the money in the present time. (2) Future

orientation is a dummy that shows that an individual selects all the choices for the future time period. However, orientations towards the future or the present might not appear on a binary but rather on a spectrum. We therefore develop another measure, not previously used anywhere (to the best of our knowledge), where we develop an index of impatience. The index reflects the average number of times a respondent chooses the present option. All these measures are then standardized to a mean of zero and a standard deviation of 1 such that they are easily interpretable in terms of standard deviations.

Finally, two more psychosocial indicators are studied. These are life expectations and happiness. To assess life expectations, the survey asked respondents if they thought their life would be better in 1 year, 3 years, and 5 years. We combine and average these three to compute a mean expectation that shows overall future expectations. The question of cash transfers and life satisfaction is increasingly being studied, showing positive causal links between cash transfers and happiness ([McGuire, Kaiser, & Bach-Mortensen, 2022](#); [Natali, Handa, Peterman, Seidenfeld, & Tembo, 2018](#)). It elicitation is a very simple binary response question asking an individual was happy. In general, we hypothesize that these psychosocial outcomes are likely to mediate welfare and investment outcomes. Following the theory of change, we predict that psychosocial outcomes mediate the effect of the intervention on primary outcomes.

3.3 Empirical Strategy

Given that the sample was drawn from individuals just around the cut-off of the PMT scores, we have a simple distribution that mimics perfect randomisation when we control (weight) the each observation by their PMT score. Our empirical

Furthermore, weighting the baseline controls by PMT scores shows that the baseline controls and the outcomes are fully balanced so that any observed effects at the endline are associated with treatment. The results of the baseline controls are shown in the Supplementary Table [A4](#). Given that the baseline was conducted before the cash transfers were implemented, we can implement Ordinary Least Squares (OLS) regressions to assess the impact of the intervention.

3.3.1 Intent to Treat Effects

While it was expected that at the end line all individuals assigned to the treatment group would have received the cash transfers and also enrolled in insurance, the enrollment in the insurance increased by only 14-15 percentage points (Palermo et al., 2019). This imperfect adherence to treatment therefore provides us with an opportunity to start evaluating the impacts through an intent-to-treat assessment. Here, we are simply concerned with the impact of being assigned to the treatment group regardless of whether the recipient fully complied with the treatment package (cash transfer + insurance) or not. We therefore estimate the following panel regression.

$$Y_{ijt} = \beta_0 + LEAP_{ijt}\beta_1 + \hat{X}_{ijt}\beta_2 + \varsigma_j + \varepsilon_{ijt}. \quad (1)$$

Where Y_{ijt} is the main outcome of interest of woman i living in community j at a point in time t measured as described above. We have several outcomes related to productive livelihoods such as livestock ownership (cows, goats, sheep, poultry), non-farm enterprise (petty trader, crafts, and other local economic activities) and livestock commercialization. We also have some psychosocial outcomes that include time preferences, happiness, expectations of the future, stress, social support, and agency. $LEAP_{ijt}$ is a dummy that takes the value 1 if the household was assigned to the program and 0 otherwise. The coefficient of interest is β_2 , which shows the effect of LEAP assignment. $\hat{X}_{ijt}$ is a vector of control variables that are used to improve the precision of the model. It refers to other individual and household characteristics as well as the PMT scores of the households. We include controls such as age and educational level, household size, marital status, and PMT scores of the households. ς_j refers to time-invariant unobserved heterogeneity (it captures fixed effects at the individual, household, and community level) and ε_{ijt} is the stochastic error term. We cluster our standard errors at the community level.

3.3.2 Local Average Treatment Effects

As mentioned above, insurance enrollment was less than expected (Otieno et al., 2022; Palermo et al., 2019). Although participation in LEAP 1000 automatically granted individuals with health insurance premium waivers, individuals were still required to enroll or renew their membership in insurance (at no cost). At the end of the intervention, it was found that many households did not activate their insurance status in part because they were not fully aware of

this additional benefit or because the travel time and costs to enroll centers were prohibitive (LEAP 1000 Evaluation Team, 2018). In addition, some households assigned to the treatment group also did not receive the cash transfer. At endline, about 88% of them were fully in compliance with treatment (Palermo et al., 2019). In general, adherence to the treatment package was imperfect, implying that the ITT results would probably be underestimated. In addition, the ITT results would not show heterogeneity within the treatment group based on how fully or partially they adhered to the treatment. Therefore, a local average treatment effect would be more precise. To estimate the local average treatment effects (LATE), our initial treatment group is now divided into two treatments reflecting their actual treatment experience. Treatment 1 corresponds to the actual adherence to the full treatment package of cash transfer and insurance, and treatment 2 corresponds only to cash transfers without insurance. The LATE equation is given by the following panel regression.

$$Y_{ijt} = \tau^{GPS} + \text{Cash \& Insurance}_{ijt}\beta_4 + \text{Cash only}_{ijt}\beta_5 + \vec{X}_{ijt}\beta_2 + \varsigma_j + \varepsilon_{ijt}. \quad (2)$$

Our coefficients of interest are β_4 which gives the coefficient for taking the full treatment package, cash and insurance, and β_5 , which gives the coefficient for taking only the cash-only component. However, note that selection into each of these treatments might be endogenously self-selected, implying that our results would be biased. This potential endogeneity cannot be ruled out using the PMT score as in the ITT analysis. Therefore, to control for this, we adopt a weighted least-squares panel regression. Given that our treatment is multivalued, we first implement a multinomial logit prediction model and estimate a generalized propensity score (Cattaneo, 2010; Uysal, 2015). We then generate weights equal to the inverse probability of the generalized propensity scores. Equation 2 is therefore a weighted least squares panel regression with weights entering the model as τ^{GPS} . Supplementary Table A6 shows the baseline controls after applying the generalized propensity score weights and the results show that almost all baseline controls are balanced and the full comparisons show no statistically significant differences in the controls.

3.3.3 Pathways

To assess the pathways of impact, we interact each psychological outcome with the treatment dummies. We estimate the following regression for each pathway (that is, impatience, future orientations, present orientations, life expectations, happiness, and agency through an agency scale)

$$Y_{ijt} = \tau^{gps} + \text{Cash \& Insurance}_{ijt} * \text{Path}_{ijt}\theta_1 + \text{Cash only}_{ijt} * \text{Path}_{ijt}\theta_2 + \vec{X}_{ijt}\beta_2 + \text{Path}_{ijt}\beta_6 + \varsigma_j + \varepsilon_{ijt}. \quad (3)$$

From the above regression, our coefficients of interest are θ_1 and θ_2 , which represent the interaction between cash plus insurance treatment and cash-only treatment and the path.

3.3.4 Multiple Hypothesis

We test a large number of outcomes, including four outcomes on livestock investments, five outcomes on non-farm investments, and six pathways (secondary outcomes). We therefore would like to rule out the potential of false discovery through multiple hypothesis. To remove these concerns, we implement two strategies for multiple hypothesis testing. First, following [Anderson \(2008\)](#), we develop three indices to represent the three groups of outcomes. Our indices are standardized inverse-covariance indices weighted by the mean of the control group ([Schwab, Janzen, Magnan, & Thompson, 2020](#)). We then implement our analysis for both the ITT and the LATE with these indices. Secondly, for each family/group of outcomes, we compute the [Benjamini, Krieger, and Yekutieli \(2006\)](#) sharpened q-values based on the [Anderson \(2008\)](#) method. The sharpened q-values are included in the main ITT and LATE result.

4 Main Results

4.1 Intention to Treat Effect

First, we look at the intent-to-treat effects. We begin with investments in livestock in [Table 1](#) below. Models 1 through 4 show the results of the livestock investments. We observe that between baseline and the end line, the probability of owning livestock among individuals assigned to LEAP 1000 increased by 7.9 percentage points or approximately 15 percent of

the baseline mean (before treatment). Livestock commercialization (buying and selling) was only observed at the endline, therefore we present cross-sectional results. We observed positive associations between assignment to the treatment group at baseline and buying and selling, but the results were not statistically significant. The Supplementary Table A7 shows heterogeneous investments in livestock of various kinds. Here, we observe that assignment to the program led to an 8.2 percentage point increase in goat ownership and a 9.1 percentage point increase in the probability of poultry ownership. We divided all the livestock into small ruminants or large ruminants¹ and observed that the effect of assignment to LEAP 1000 was dominant by small ruminant effects, a close to 10 percentage points increase in the probability of livestock ownership.

Table 1: Intention to Treat Effects of LEAP 1000 on household investments

VARIABLES	(1) Livestock	(2) TLU	(3) Buying	(4) Selling	(5) Durable goods	(6) Business	(7) Savings	(8) Loan	(9) Non farm
Treatment	0.079*** (0.022)	-0.096 (0.182)	0.039 (0.036)	0.003 (0.021)	-0.006 (0.070)	0.016 (0.016)	0.214*** (0.022)	0.040 (0.026)	0.117*** (0.026)
undjusted p-values	0.000	0.598	0.271	0.880	0.928	0.320	0.000	0.125	0.000
Sharpened q-values	0.001	0.427	1.000	1.000	0.591	0.265	0.001	0.144	0.001
Constant	0.215* (0.122)	0.262 (0.680)	0.218*** (0.069)	0.012 (0.055)	6.311*** (0.314)	0.155 (0.106)	0.108 (0.107)	0.095 (0.128)	0.072 (0.128)
	0.080	0.700	0.002	0.833	0.000	0.144	0.312	0.459	0.572
Observations	4,658	4,658	2,329	2,329	4,658	4,658	4,536	4,658	4,658
R-squared	0.607	0.597	0.081	0.036	0.648	0.611	0.609	0.602	0.574
Baseline mean	0.517	1.506			567.5	0.213	0.0785	0.482	0.279

All regressions control for the variables included in Table A4, community fixed effects and language fixed effects. Significance levels correspond with *** p<0.01, ** p<0.05, * p<0.1

Models 5 through 9 in Table 1 show results of non-farm investments. The assignment to the program had a positive and significant effect on the savings. The probability of having some savings increased by 21.4 percentage points (7. 9%). Also, the probability of taking on non-farm employment by at least one household member increases by about 12 percentage points. Note that all significant ITT effects are robust to multiple hypothesis testing.

Table 2 shows the intent-to-treat results of the effect of assignment to the intervention. We find that assignment to the treatment group increased impatience by about 0.1 standard deviations while having almost the same margin of effect on future orientations, in the opposite direction - reducing them by about 0.1 standard deviations. The measure of impatience is an index that comes from a combination of the six questions in the time preference elicitation.

¹Large ruminants were either cattle or donkey while small ruminants were goats, sheep and pigs. Technically, pigs are not ruminating animals and therefore are their own category of pigs/swine. However, for computational simplicity, we included pigs in the small ruminant category. Their inclusion does not change the inference of the results in any way

However, the present orientation dummy shows if an individual always selects the present offer rather than a future offer. Looking at the present orientations, we observe that assignment to the intervention increases the present orientations by slightly more than 0.07 standard deviations. Model 4 looks at the effect of assignment to the intervention on happiness. We observe that the intervention increases happiness by about 0.21 standard deviations. Model 5 shows the results of expectations for the future. We find that assignment to the treatment group increases expectations for the future by 0.16 standard deviations. The last result in Table 2 shows the results of the women's agency. We observe that overall agency improves by 0.37 standard deviations as a result of assignment to the intervention. Again, we note that all the results in Table 2 are robust to multiple hypothesis testing.

Table 2: Intent-to-treat results of the effect of LEAP 1000 on psychosocial outcomes

VARIABLES	(1) Impatience	(2) Future orient	(3) Present orient	(4) Happiness	(5) Expectation	(6) Agency
LEAP 1000	0.105* (0.055)	-0.099* (0.054)	0.072*** (0.026)	0.206*** (0.045)	0.164*** (0.055)	0.373*** (0.059)
unadjusted p-values	0.058	0.070	0.006	0.000	0.003	0.000
Sharpened q-values	0.024	0.025	0.006	0.001	0.005	0.001
Constant	-0.670** (0.311)	0.682** (0.311)	0.109 (0.145)	-0.136 (0.248)	0.076 (0.288)	0.394 (0.289)
	0.033	0.030	0.451	0.584	0.791	0.175
Observations	4,536	4,536	4,536	4,658	4,536	4,536
R-squared	0.545	0.545	0.539	0.563	0.545	0.587

All regressions control for the variables included in Table A4, community fixed effects and language fixed effects. Robust standard errors in parenthesis. Significance levels correspond with *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

4.2 Local Average Treatment Effects

Table 3 below shows the results of Equation 2. Models 1 to 4 show results on livestock investments. First, the main result is that cash plus insurance and cash-only treatments generally have different results, with the key result dominated by the cash plus insurance treatment. We observe that individuals who took the full treatment package (cash plus health insurance) had a higher probability of having livestock. In this group, the probability of having livestock increased by 7.7 percentage points, whereas in the cash-only group, the result is not only insignificant. We do not observe any significant results on total livestock units. Because livestock commercialization was measured only in the end-line, we present cross-sectional results only from the end-line. Looking at these results, we observe a similar trend. Individuals

taking the full treatment package had a 9 percentage point increase in the probability of purchasing livestock, while those who took cash only seemed to reduce their probability of purchasing livestock by approximately 8 percentage points. However, both these results were not robust to multiple hypothesis testing. We did not observe any significant effect on the probability of livestock sales, potentially indicating that both treatments had a positive effect on assets accumulation while also protecting households from asset depletion.

Models 5 through 9 show the results of non-agricultural household investments. Starting with durable goods, we do not observe any significant causal relationship in either of the two treatments. We also do not observe any significant relationship regarding starting a business and taking a lot of, however, we observe that being just exposed to the cash plus insurance treatment have a 22 percentage point increase in the probability of having some service whereas those exposed to the cash only treatment have a 28 percentage point increase in the probability having some savings. The finding that cash-only treatment had a larger effect on savings potentially aligns with the costs of accessing insurance, especially transport costs, that could reduce savings. Putting these results in their magnitude change from the baseline mean cash plus insurance increases the probability of having some savings by about 2.8 times, whereas cash only treatment increases the probability of having some savings by about 3.5 times.

Finally, Model 9 shows results of at least one individual in the household taking a non-farm wage job, and here we observe that cash plus insurance treatment increases the probability of having a non-farm wage job by about 11.7 percentage points, equivalent to about 42% of the best-line mean. The coefficient for cash-only treatment is also positive; it is very small and is not close to any levels of conventional statistical significance.

Supplementary Table [A8](#) shows LATE heterogeneity by the type of livestock households that are more likely to invest in by running Equation [2](#) with the respective livestock dummies. Our findings indicate that, like the main LATE effects, the treatment effect was dominated by the complete treatment package compared to those who took only the cash transfer. We do not observe a significant effect relating to cattle and instead find that receiving both insurance and the cash transfer increased the probability of investing in goats by close to 10 percentage points and in poultry by approximately 8.3 percentage points. We further divided the livestock into two groups, namely; large ruminant animals and small ruminant animals. We found that receiving only cash was casually associated with disinvestment in large ruminants by about 3.2

Table 3: Local Average Treatment Effects of LEAP 1000 on household investments

VARIABLES	(1) Livestock	(2) TLU	(3) Buying	(4) Selling	(5) Durable goods	(6) Business	(7) Savings	(8) Loan	(9) Non-farm job
Cash + Insurance	0.077** (0.033)	0.305 (0.301)	0.091** (0.045)	0.018 (0.025)	-0.070 (0.108)	0.021 (0.022)	0.219*** (0.031)	0.033 (0.031)	0.117*** (0.032)
unadjusted p-values	0.020	0.312	0.044	0.467	0.521	0.341	0.000	0.297	0.000
Sharpened q-values	0.086	0.745	0.166	0.453	0.807	0.660	0.001	0.660	0.002
Cash only	-0.055 (0.090)	-0.224 (0.393)	-0.078* (0.043)	-0.010 (0.028)	0.115 (0.257)	-0.001 (0.021)	0.278*** (0.084)	-0.118 (0.087)	0.025 (0.098)
unadjusted p-values	0.541	0.569	0.071	0.717	0.655	0.974	0.001	0.174	0.800
Sharpened q-values	0.745	0.745	0.166	0.559	0.965	1.000	0.004	0.437	1.000
Constant	0.274** (0.121)	1.102 (0.897)	0.295*** (0.088)	0.031 (0.054)	6.396*** (0.407)	0.086 (0.131)	0.014 (0.124)	0.116 (0.140)	-0.001 (0.136)
	0.024	0.221	0.001	0.568	0.000	0.512	0.912	0.408	0.994
Observations	4,658	4,658	2,329	2,329	4,658	4,658	4,536	4,658	4,658
R-squared	0.619	0.600	0.088	0.039	0.643	0.636	0.622	0.618	0.576
Baseline mean	0.517	1.506			567.5	0.213	0.0785	0.482	0.279

All regressions control for the variables included in Table A4, community fixed effects and language fixed effects. Robust standard errors in parenthesis. Significance levels correspond with*** p<0.01, ** p<0.05, * p<0.1

percentage points, while cash + insurance did not statistically significantly change the trajectory of large ruminant investments. However, for small ruminant animals, we observed that cash plus insurance led to a 13.2 percentage point increase in the probability of investment in them.

Lastly, we examine the effect of the intervention on psychosocial outcomes. Table 4 below shows the effect of these outcomes. Models 1 to 3 are derived from the time preference dissertation module and show impatience (Model 1) future orientations (Model 2) or present orientations (Model 3). We observe that receiving cash plus insurance increases impatience by about 0.21 standard deviations, whereas cash only does not show a statistically significant effect. Reversing the way in which we compute and consider time preferences, we look at the effect of the intervention on future orientations and observe that receiving cash plus insurance reduces future orientations by approximately 0.17 standard deviations, while cash alone does not affect future orientations. Model 3 looks at present orientations where, in each of the six time preference questions, the respondent would have chosen the present rather than the future time period. Here, we observe that receiving the cash plus insurance increases present orientation by about 0.094 standard deviations, while cash only does not have a statistically significant effect. The key message regarding time preferences is that station here is that cash plus insurance increases present orientation, thereby impatience while reducing future orientation. In other words, treated individuals prefer to have the benefit now rather than wait for the future.

Table 4: Local Average Treatment Effects of LEAP 1000 on psychosocial outcomes

VARIABLES	(1) Impatience	(2) Future orient	(3) Present orient	(4) Happiness	(5) Expectation	(6) Agency
Cash + Insurance	0.209*** (0.066)	-0.165** (0.066)	0.094*** (0.032)	0.170*** (0.056)	0.137** (0.064)	0.353*** (0.084)
unadjusted p-values	0.002	0.014	0.004	0.003	0.033	0.000
Sharpened q-values	0.011	0.017	0.011	0.011	0.029	0.001
Cash only	0.022 (0.077)	-0.017 (0.076)	0.013 (0.041)	0.149** (0.063)	0.101 (0.079)	0.770*** (0.269)
unadjusted p-values	0.776	0.823	0.748	0.018	0.204	0.005
Sharpened q-values	0.375	0.375	0.375	0.019	1.000	0.011
Constant	-0.804*** (0.307)	0.787** (0.314)	0.109 (0.150)	-0.053 (0.282)	-0.266 (0.312)	0.228 (0.393)
	0.010	0.013	0.471	0.851	0.395	0.563
Observations	4,536	4,536	4,536	4,658	4,536	4,536
R-squared	0.563	0.567	0.557	0.570	0.556	0.596

All regressions control for the variables included in Table A4, community fixed effects and language fixed effects. Robust standard errors in parenthesis. Significance levels correspond with *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Model 4 looks at the effect of the intervention on happiness. Here, we find that both treatments have a positive effect on happiness with cash plus insurance increasing happiness by about 0.17 standard deviations while cash only increases it by about 0.15 standard deviations. It is expected that the intervention also causally influences people's expectations of how the future might turn out. Combining the three questions about expectations about the future into one expectation index, Model 5 looks shows that both treatments increase individuals' assessment of future expectations. Cash plus insurance increases people's expectations by about 0.14 standard deviations whereas cash only though with a positive coefficient is not statistically significantly affecting future expectations. Agency is measured by a set of six questions that establish individuals' self-efficacy and sense of control. Model 6 shows these results, revealing that cash provided a higher sense of agency than cash plus insurance. Among individuals in cash plus insurance, the agency increases by 0.35 standard deviations, while among individuals receiving cash only, it increased by more than twice as much, about 0.77 standard deviations. All our results here are robust to multiple hypothesis.

4.3 Pathways: Psychosocial outcomes and productive investments

Following our theory of change, we predict that psychosocial outcomes affect household investments by how they could positively modify planning and resource allocation. Thus, we implement Equation 3 for each of the psychosocial outcomes we study. The panel regressions are also weighted by the inverse probability of the generalized propensity score. We focus on the interaction between the pathway and the treatment groups.

Effect of Impatience and Present Orientations: Table 5 shows the mediation effect of time preferences and future expectations on household investments. Panel 1 of Table 5 shows the effect of impatience. We observe that impatience has a statistically significant causal relationship with about three of the nine outcomes in household investments. First, we note that among livestock investments, impatience statistically significantly affects livestock purchase by reducing the probability of livestock purchase by about 6.6 percentage points among those who received the full treatment package. For those who took only the cash transfer component, impatience increases the purchase of livestock by about 5.3 percentage points. We also observe that impatience was casually related to investments in durable goods. Among those who receive both insurance and cash transfers, an increasing level of impatience was casually linked with reducing durable investments by 11 percentage points and for those receiving only the cash transfer, impatient individuals reduced durable goods investments by an even greater margin of about 36 percentage points. Notice that only treatments, without considering the effect of impatience, do not change recipients' investments in durable assets (see Table 3 above). Finally, impatience reduced the probability of savings by about 4.7 percentage points, significant only among those taking the full treatment package.

Effect of Future Orientations: Panel 3 of Table 5 shows the effect of future orientations. We observed that impatience generally has a negative effect on investments in households, future orientations have a positive effect on investment. In a way, future orientations are a direct mirror image of impatience. We observe that a more comprehensive treatment (cash transfer and insurance) led to an increase in livestock purchase by 5.8 percentage points but a more reduced treatment (cash only) reduced livestock purchase by about 5.2 percentage points. We also observe that future orientations increased investment in durable goods. The combination of cash transfers and insurance led to an increase in durable goods by approximately 11 percentage

Table 5: Heterogeneity of livestock and non-farm investments by dimension of psycho-social wellbeing

VARIABLES	(1) Livestock	(2) TLU	(3) Durables	(4) Business	(5) Savings	(6) Loans	(7) Non farm
Panel 1: Impatience							
Cash + insurance x impatience	-0.001 (0.027)	-0.185 (0.249)	-0.150** (0.058)	0.025 (0.023)	-0.031 (0.023)	-0.043 (0.032)	-0.002 (0.027)
Cash only x impatience	0.003 (0.062)	0.534 (0.566)	-0.380*** (0.138)	0.011 (0.027)	-0.031 (0.055)	0.033 (0.060)	0.084 (0.066)
Panel 2: Future Orientations							
Cash + insurance x future orientations	-0.000 (0.026)	0.205 (0.253)	0.143** (0.059)	-0.025 (0.022)	0.026 (0.023)	0.043 (0.032)	0.000 (0.027)
Cash only x future orientations	-0.007 (0.066)	-0.416 (0.623)	0.395*** (0.142)	-0.006 (0.027)	0.030 (0.060)	-0.028 (0.063)	-0.107 (0.069)
Panel 3: Expectations about the Future							
Cash + Insurance x expectations	0.074 (0.141)	3.759* (2.136)	0.270 (0.423)	0.237** (0.112)	0.226 (0.141)	-0.023 (0.154)	-0.160 (0.164)
Cash only x expectations	-0.265 (0.224)	0.694 (2.501)	0.091 (0.586)	0.138 (0.126)	0.239 (0.208)	-0.480** (0.239)	-0.754*** (0.229)
Panel 4: Happiness							
Cash + insurance x happiness	-0.026 (0.026)	-0.174 (0.115)	0.157 (0.099)	0.063*** (0.023)	0.084*** (0.022)	-0.014 (0.025)	-0.090*** (0.025)
Cash only + happiness	-0.065 (0.050)	-0.364* (0.191)	0.345** (0.156)	-0.006 (0.043)	0.059 (0.053)	-0.048 (0.053)	-0.056 (0.060)
Panel 4: Agency/ Control of one's life							
Cash + insurance x agency	0.002 (0.029)	0.237 (0.238)	-0.227*** (0.071)	0.019 (0.023)	0.067** (0.026)	0.056 (0.034)	0.045 (0.027)
Cash only x agency	-0.024 (0.064)	-0.224 (0.627)	0.175 (0.134)	-0.054 (0.045)	0.005 (0.053)	-0.011 (0.062)	0.040 (0.070)

All regressions control for the variables included in Table A4, community fixed effects and language fixed effects. Robust standard errors in parenthesis. Significance levels correspond with *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

points. However, cash only led to an increase in durable goods by 37 percentage points when combined with future-oriented behavior. In addition, cash plus insurance combined with future-oriented behavior led to an increase in savings by about 4.2 percent.

Effect of Present Orientations: Panel 2 of Table 5 shows the effect of present orientations. The results are similar to those of impatience, however, with an even larger magnitude. For example, individuals with present orientations, when combined with cash and insurance, reduced the probability of purchasing livestock by 12 percentage points. However, present orientations combined with cash-only treatment increased the probability of livestock purchase by about the same magnitude (12 percentage points). We also observe that the more comprehensive treatment package combined by present orientations reduced the probability of savings by 11 percentage points and also reduced the probability of taking loans by 10 percentage points.

Effect of Expectations about the Future: Panel 4 shows the mediating effect of expectations about the future. From a livestock investment perspective, individuals with better future expectations who received the full intervention package increased their total livestock unity by more than 2.5 times (baseline mean 1.5 TLU). The result is significant only at a 10 percent level. Those with only the cash transfer, but also with better future expectations also had a positive outcome, although not at all significant. As individuals with better expectations for the future increased their livestock holdings (TLUs), they also reduced the probability of selling. This was especially true for those who had cash only intervention, whose probability of selling livestock was reduced by more than 28 percentage points.

Effect of Happiness: Table 6 shows the mediating effect of other psychosocial factors, namely; happiness, social support, stress, and agency. Panel 1 of Table 6 shows the results of happiness. We observe that happiness was causally related to reductions in livestock investments, especially in livestock units and purchase. However, this was only statistically significant for individuals receiving only cash transfer, for whom increasing happiness led to a 0.36 reduction in total livestock units and a 6.3 percentage point reduction in livestock purchase. Regarding non-farm investments, we saw that the mediating effects were

Table 6: Mediating Effect of Happiness and Agency

VARIABLES	(1) Livestock	(2) TLU	(3) Buying	(4) Selling	(5) Durables	(6) Business	(7) Savings	(8) Loans	(9) Non farm
Happiness									
Cash + Insurance x Happiness	-0.026 (0.026)	-0.168 (0.114)	0.021 (0.019)	-0.007 (0.017)	0.170* (0.101)	0.064*** (0.023)	0.078*** (0.022)	-0.017 (0.024)	-0.097*** (0.025)
Cash only x Happiness	-0.065 (0.049)	-0.357* (0.201)	-0.063** (0.032)	-0.009 (0.018)	0.360** (0.153)	-0.005 (0.043)	0.052 (0.052)	-0.052 (0.052)	-0.063 (0.060)
Constant	0.277** (0.120)	1.115 (0.898)	-0.214 (0.131)	-0.114* (0.063)	6.385*** (0.405)	0.086 (0.131)	0.015 (0.121)	0.117 (0.140)	0.000 (0.135)
Observations	4,658	4,658	4,658	4,658	4,658	4,658	4,536	4,658	4,658
R-squared	0.620	0.600	0.549	0.523	0.648	0.637	0.625	0.619	0.579
Agency									
Cash + Insurance x Agency	0.003 (0.029)	0.234 (0.236)	-0.034 (0.025)	0.045*** (0.016)	-0.199*** (0.072)	0.021 (0.024)	0.058** (0.026)	0.051 (0.035)	0.034 (0.028)
Cash only x Agency	-0.020 (0.067)	-0.231 (0.606)	-0.068** (0.033)	-0.031 (0.022)	0.250* (0.129)	-0.048 (0.045)	-0.021 (0.056)	-0.023 (0.064)	0.011 (0.072)
Constant	0.159 (0.133)	1.188 (1.098)	-0.242 (0.153)	-0.182** (0.076)	6.252*** (0.413)	0.019 (0.138)	-0.018 (0.124)	0.150 (0.143)	-0.007 (0.139)
Observations	4,536	4,536	4,536	4,536	4,536	4,536	4,536	4,536	4,536
R-squared	0.623	0.607	0.549	0.527	0.651	0.637	0.624	0.618	0.576
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

All regressions control for the variables included in Table A4, community fixed effects and language fixed effects. Robust standard errors in parenthesis. Significance levels correspond with *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Effect of Agency: Finally, Panel 2 of Table 6 shows the results of agency measured by an agency index. Here, we observe that in households with higher levels of agency and receiving the full treatment package, the probability of selling livestock increased by about 4.5 percentage points, whereas those who only receive cash and also have high levels of agency reduced livestock purchase by about 6.8 percentage points. An increase in levels of agency combined with cash plus insurance reduce investments in durable assets by about 20 percentage points, but for those who only receive cash instead the effect is the opposite, increasing their probability of investment in durable goods by about 25 percentage points. We also observe that the full treatment package, when combined with increasing agency, leads to a 6 percentage point increase in the probability of having some savings.

5 Discussion

In general, we observe that the LEAP1000 program positively influenced livestock ownership and commercialization. In addition, the positive effects on livestock investments are largely limited to small ruminant animals such as goats and sheep, and poultry. This is an intuitively expected finding given the size of the transfers. Whereas the transfer enables some substantial

investments, it might not be large enough to significantly shift investment in expensive and more costly to rear, large ruminant animal such as cattle. Beyond this, it could also be the case that goats and poultry are easily commercialized and therefore preferred by households. In addition, these categories of livestock, especially poultry, are easy to manage as they require minimal attention, and households can participate in other profitable employment options when investing in these livestock. In addition, small-ruminant livestock production is in line with the small-scale agricultural production systems of rural households. since most poultry are usually produced on farms. For example, in Tanzania, poultry production is well integrated in vegetable production in the form of a feedback loop ([Habiyaemye, Tabe-Ojong, Ochieng, & Chagomoka, 2019](#)). In these vegetable-cum poultry production systems, poultry depend on and are usually fed agricultural products from farms, and their litter, which comprises excreta and spilled feed, is used as a form of organic manure in these vegetable production systems. This kind of circular production system is less costly but also environmentally friendly, since households do not engage in the use of chemical production methods using inorganic fertilizers, which may be costly for them. It has even been established that the use of these types of organic nature of livestock is associated with food security in rural settings in Cameroon ([Tabe-Ojong, René, Nzié, & Fabinin, 2023](#)).

Our study contributes to an increasing body of literature on cash transfers and livestock investments, with findings more or less in the same direction and, to a greater extent, even of similar magnitudes ([Asfaw et al., 2014](#); [Daidone et al., 2019](#); [De Hoop, Groppo, & Handa, 2020](#); [Ibrahim et al., 2023](#); [Shigute et al., 2020](#); [Todd, Winters, & Hertz, 2010](#)). These studies assessed programs that cover large-scale transfer programs in sub-Saharan Africa (Ethiopia, Ghana, Kenya, Lesotho, Malawi, Zambia and Zimbabwe). However, a number of differences are recorded between these studies and ours. First, our study exclusively focuses on poor women as a specifically vulnerable subgroup different from the average households cash transfers usually target. In this case, we contribute a small group of studies that more specifically study effects among poor women. Secondly, apart from [Shigute et al. \(2020\)](#), none of the other studies is a "cash +" program. Our study is one of the few cash plus health insurance programs that considers these set outcomes². But our study is also consistent with regard to how different livestock respond to transfer. For example, like [Daidone et al. \(2019\)](#), in their multi-

²Other cash plus health insurance studies such as [Mussa, Agegnehu, and Nshakira-Rukundo \(2022\)](#) and [Mussa, Agegnehu, and Nshakira-Rukundo \(2024\)](#) looking at food security outcomes and is potentially underpowered due to their small samples and cross-sectional nature

country study, we also find that overall livestock units do not shift significantly and that people invest more in smaller animals compared to large animals such as cattle. Across the board, the value of the transfer seems to dictate where people allocate investments. Perhaps future studies can consider the potential differences from large one-time transfers such as [Egger, Haushofer, Miguel, Niehaus, and Walker \(2022\)](#)

With regard to psychosocial outcomes, an increasing body of research is emerging on the effects of cash transfers on psychosocial well-being and future orientations. This literature can be divided into three strands. The first strand considers working on cash transfers and aspirations ([García, Harker, & Cuartas, 2019](#); [Kosec & Mo, 2017](#); [Patel-Campillo & García, 2022](#); [Stampini, Martinez-Cordova, Insfran, & Harris, 2018](#); [Suarez & Cameron, 2020](#); [Yu, Liu, Wang, & Lin, 2024](#)). Expectations are slightly different from aspirations. Although some researchers consider aspirations as a measure of expected future income (and therefore expectations and aspirations as synonymous) ([Genicot & Ray, 2020](#)) in other situations, researchers consider these separately ([Galiani, Gertler, & Undurraga, 2021](#)). Our study measured only expectations – beliefs about the future rather than aspirations – wishes with which one is working on and hopes to achieve in a given time frame with a certain degree of confidence. In general, studies that evaluate cash transfers and aspirations tend to focus on educational aspirations of children in recipient households ([García et al., 2019](#); [Patel-Campillo & García, 2022](#); [Stampini et al., 2018](#); [Suarez & Cameron, 2020](#)). We therefore provide a more broader assessment of the general future, and not a single dimension of the future. We further argue that aspirations and expectations are further supported by (a) people’s self-restraint in present gratification while investing in actions and activities with a future orientation. The two sets of variables we use here are time preferences, which measure present orientations, future orientations, and impatience and social support.

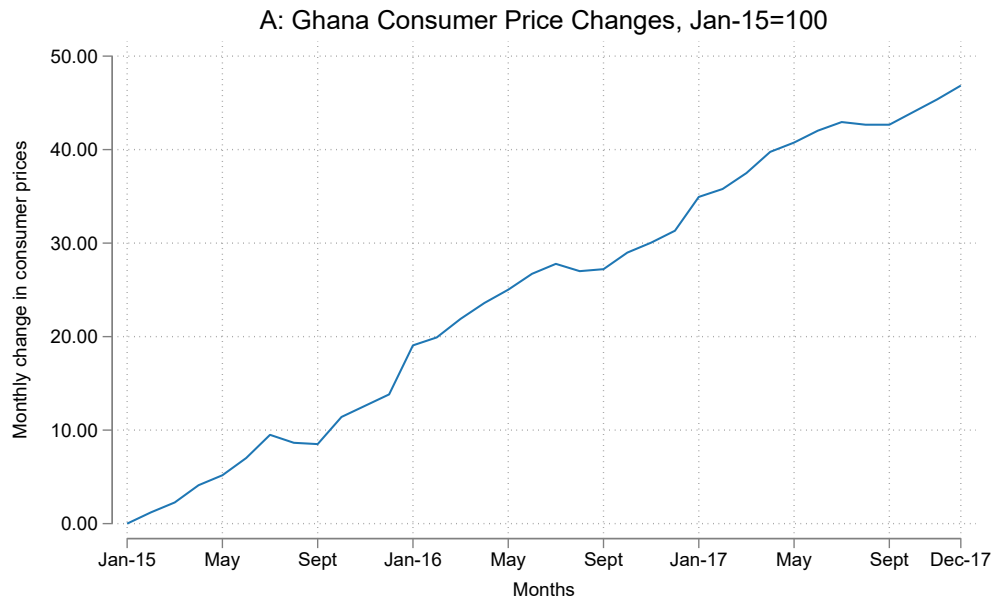
Our work on contributes to other recent studies that investigate how cash transfers affect time preferences [Handa et al. \(2014, 2020\)](#); [Ibrahim et al. \(2023\)](#); [Maara et al. \(2023\)](#); [Molotsky and Handa \(2021\)](#). [Molotsky and Handa \(2021\)](#) and [Maara et al. \(2023\)](#) study the effect of cash transfers in Malawi and Zimbabwe, respectively, on various psychosocial outcomes, including time preferences, stress, happiness, expectations for the near (1 year) future, quality of life and worry. They find that the cash transfer greatly improved these indicators. They also test how these psychological outcomes affect savings and find that all positively framed outcomes (e.g. reducing stress) were associated with increasing the probability of savings. [Angeles et](#)

al. (2019) study the effects on youth mental health and observe that cash transfers reduced depressive symptoms, as well as the general depression scale among young people. Our results agree well with these studies in outcomes such as expectations about the future, happiness and agency, but not in time preferences. We cannot explain this fully.

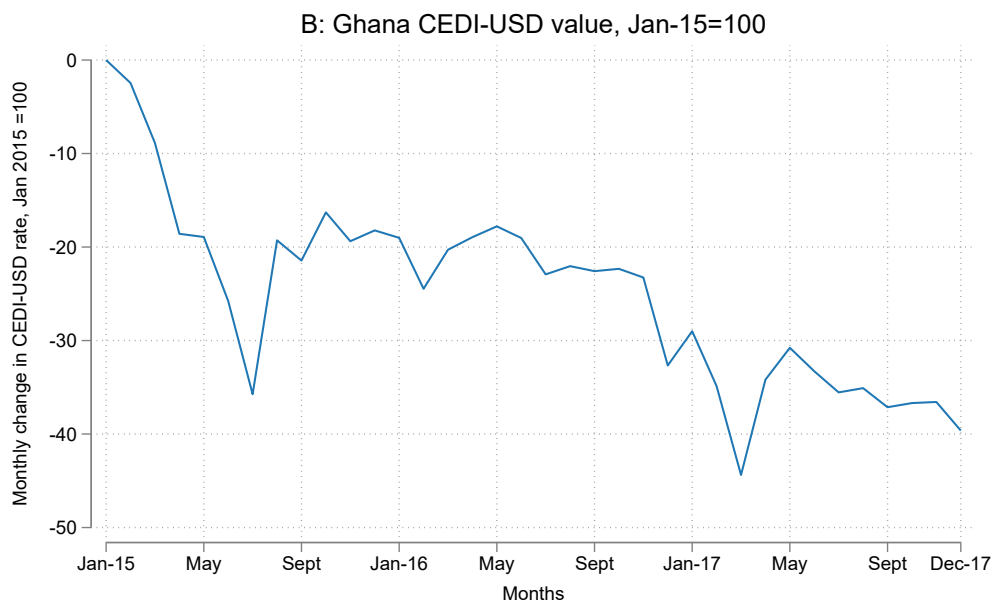
However, one hypothesis could be the simple cultural differences between countries in time preferences (De Lipsis, 2021; Wang, Rieger, & Hens, 2016). Another spatial reason could be the economic differences between countries in relation to inflation and the general value of money (Gong, 2006). In the last decade, Ghana's economic situation has been tumultuous with high inflation, an increase in consumer prices, and a loss of value of the Ghanaian Cedi. Figure 2 below shows the economic conditions in Ghana during the implementation period of LEAP 1000. Using Consumer Price Index data and scaling January 2015 as the starting point, we observe that average consumer prices increased by close to 50%. In Figure B showing the Cedi-USD exchange rates during the period, we use historical exchange rates to assess the potential loss of value of the Ghanaian Cedi and observe that between baseline and end line, the local currency lost approximately 40% of its value on the US dollar. In addition to these macro-evaluations, local price data was collected as part of the evaluation process. We selected household high-frequency food and non-food items whose prices for which prices were recorded at least 100 times in both baseline and endline combined. Our list included 21 food items and petrol. Supplementary Figure A1 shows the changes in prices between baseline and endline. We observe that approximately half of the items selected experienced price increases with the highest increase of approximately 112% experienced for powder milk, which is essential for children. Other key food items such as sugar, gari, millet, beans, and fish also had high prices. This could therefore explain why time preferences were a lot more skewed toward the present than the future.

The other potential hypothesis could be the differences in our treatments in contrast to other studies. Our paper is the first to simultaneously assess combinations of cash plus insurance or cash alone on time preferences. Perhaps of more nuance is how we observe a shift to present orientations at the same time higher or better expectations about the future. We observe that when people believe that the future will be better, they are more likely to invest in the present situation to alleviate the present need. This might show as if the cash transfer reduces future orientations, but putting it in the context of how the future might turn out makes nuanced sense that individuals might invest in the future if they believe that the future might be worse than the

Figure 2: Negative economic condition during LEAP 1000 implementation



Source: Statistics Ghana Consumer Price Index reports (2015 - 2017)



Source: Yahoo finance historical exchange rates (2015 - 2017)

present. Future orientations therefore act as a stimulant for insurance for the future. However, cash transfer can provide capital for productive investments so that future expectations improve. This might in turn reverse the spectrum of future-present orientations.

To test out theory of change regarding pathways, we assess how psychosocial outcomes affect investment decisions in livestock and non-farm activities such as seeking wage jobs or starting businesses. Generally, higher aspirations drive investments in assets such as livestock (Tabe-Ojong, Heckeleei, & Rasch, 2023) and cash transfers might enhance these decisions. Previous research by Molotsky and Handa (2021) and Ibrahim et al. (2023) established that some of these psychosocial outcomes mediated investments in livestock and savings among cash transfer recipients. A new addition to this literature therefore is the potential contributory factor of health insurance on cash transfer packages, which none of the previous studies have done. We test this a range of psychosocial indicators and find that indeed, they influence investment decisions. Impatient individuals are less likely to purchase livestock, invest in durable goods, or have savings, while the opposite is true for future-oriented individuals. We further find that expectations about the future, when combined with the cash + component, increase livestock units by more than twice as much while also protecting households from selling livestock by reducing the probability of livestock sales. This is more profound for households in the cash-only intervention. Moreover, the full treatment package interacted with expectations increased the probability of owning a business. However, people who did not receive insurance were likely to not take loans and also not work in wage jobs. In general, positive expectations for the future were transformative and protective, when combined with both insurance and transfers.

Research on the economics and economic impact of happiness, especially in relation to cash transfers (McGuire et al., 2022). Although happiness can mediate some aspects of development, it can also inhibit others, for example, reducing time preferences and inducing impatience among people (Ifcher & Zarghamee, 2011). Molotsky and Handa (2021) and Ibrahim et al. (2023) find that happiness increases investments through its positive effect on time preferences, livestock holding, and savings. Our results are much more mixed, between treatments and between outcomes. Livestock-related outcomes appear to be negatively affected as a result of cash-only, while happiness positively mediated durable investments, savings, and business ownership among households with the full treatment package. However, we also observed that among households receiving the comprehensive cash + treatment, there was a

close to 10 percentage points lower probability of having a member in wage employment. Our headline result shows that cash + insurance leads to a 12 percentage point increase in wage employment, which aligns with the conventional knowledge that cash transfers provide individuals with income to facilitate job search and do not crowd out labour market participation [Baird, McKenzie, and Özler \(2018\)](#); [Handa et al. \(2018\)](#). But future expectations and happiness pathways do not seem to mediate this effect. In some cases, happiness and employment are not related ([Gielen & van Ours, 2014](#)). Happiness and other positive affect indicators can also be short-lived ([Galiani, Gertler, & Undurraga, 2018](#); [Galiani et al., 2021](#)). Therefore, such a result as we observe might not be as unexpected as it might seem. Another potential reason could be that getting people into formal employment potentially requires more than what the cash transfer provides. First, it is likely that people who took the full treatment package were poorer than those who took cash only treatment, even when these differences might not be statistically significant. In the supplementary figure [A2](#), we show the distribution of PMT scores for the two treatments and the control. We observe that the distribution for cash + insurance was slightly different (shorter bell) from that of cash-only treatment, which could imply more experience of poverty.

Finally, both treatments increase agency in both the ITT and LATE estimates. The mitigation capacity of the agency was much more mixed. When combined with cash only, it reduced the probability of buying livestock, and when combined with both cash and insurance, it increased the probability of selling livestock. Agency also had mixed effect on the purchase of cattle depending on which treatment it was. Among the full treatment package, the agency led to a lower investment in durable goods of about 20 percent and also increased the probability of having some savings by approximately 6 percentage points. However, among the cash-only treatment, the agency increased investment in durable goods by about 25 percent. The potential interpretation here could be that when individuals have comprehensive social protection and also feel more in control of their lives, they could be more likely to invest in the future (savings) than tie their resources in non-moveable and potentially less liquid assets such as durable goods.

6 Conclusion

We study the effect of a cash + intervention that combines health insurance and cash transfers on economic and psychosocial outcomes in Ghana. We rely on a two-wave panel data set and leverage a thin cut-off of a proxy means test that assigns treatment. Our sample is taken from both sides of the proxy means test, but it is very close to the threshold on both sides such that there are no significant differences in observed characteristics when the sample is conditioned on the proxy means test. The treatment was a cash transfer and a waiver of a fee for health insurance. However, after assignment to treatment, there were differences in adherence to treatment. We are therefore able to evaluate both the intent-to-treat effects from only the assignment and the local average treatment effect. The latter separates the treatment group into full adherence to treatment (that is, receiving the cash transfer and also utilizing insurance premium waivers) or partial adherence to treatment through receiving only the cash transfer. Following the intervention theory of change, we study effects on livestock investments, non-farm investments, and psychosocial outcomes. Our intent-to-treat results show that the intervention increased the probability of livestock ownership, increased the probability of having some savings, and also increased the probability of household members taking on wage jobs. We find that the intervention increased happiness, expectations for the future and women's agency. However, it increased present orientations and impatience.

Our results of the local average treatment effect show that adhering to the full treatment package made the key difference. All increases in the probability of livestock ownership and wage jobs accrued to cash plus insurance treatment, while the probabilities of livestock purchase and selling were higher among the cash plus insurance group compared to the cash-only group. Similarly, the effects of the intervention on psychosocial outcomes were dominated by cash + insurance treatment. All time preferences effects (impatience, future orientations and present orientations) were significant only in the cash plus insurance treatment. We find that the effects on happiness and expectations for the future were larger among this treatment as well. Providing cash only had greater effects on women's agency than when cash was combined with insurance.

Our results largely align with other recent cash plus studies such as [Shigute et al. \(2020\)](#) and [Bossuroy et al. \(2022\)](#) on the strong evidence of cash plus interventions. Closer to our study, we contribute to recent studies that have assessed the broader impacts of the LEAP 1000 intervention in Ghana, including intimate partner violence ([Peterman et al., 2022](#)), enrolment

in health insurance (Otieno et al., 2022; Palermo et al., 2019), child nutrition (de Groot et al., 2022), health outcomes (Novignon et al., 2022) including birth weight (LaPointe et al., 2024; Quinones, Lin, et al., 2023; Quinones, Mendola, et al., 2023), community social support (de Milliano, Barrington, Angeles, & Gbedemah, 2021), and stress (Maara et al., 2023). A summary of these evaluation studies emanating from the LEAP 1000 program is presented in Supplementary Table A3. However, none of these studies assesses the implications of the intervention on household investment choices and participation in economic activities. In addition, almost all of them assess only the intent-to-treat effects and do not assess the more precise local average treatment effect based on actual treatment experience.

The insights of this study are very informative and relevant for developing countries, where high levels of poverty are usually compounded by low formal and wage employment rates, poor nutrition, and low agricultural productivity among the myriad of development challenges. Although cash transfers have proven important in the last few decades, recent evidence, including what is presented here, shows that cash plus interventions have larger effects and therefore should be prioritized.

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Appendix

Table A1: Amount of cash transfer per household

Household size	Transfer amount in Ghana ¢
One eligible member	64
Two eligible members	76
Three eligible members	88
Four eligible members	106

Source: [Transfer Project \(2020\)](#)

Table A2: Mode of elicitation of time preferences

Suppose someone you trust gives you some money. You can choose to receive GH¢100 now or an amount at a later date. What would you choose? Select the first option or the second option		
1	GH¢100 now or GH¢100 in one month	1= 100 GH¢Now 2= 100 GH¢in 1 month
2	GH¢100 now or GH¢300 in one month	1= 100 GH¢Now 2= 300 GH¢in 1 month
3	GH¢100 now or GH¢75 in one month	1= 100 GH¢Now 2= 75 GH¢in 1 month
4	GH¢100 now or GH¢200 in one month	1= 100 GH¢Now 2= 200 GH¢in 1 month
5	GH¢100 now or GH¢150 in one month	1= 100 GH¢Now 2= 150 GH¢in 1 month
6	GH¢100 now or GH ¢250 in one month	1= 100 GH¢Now 2= 250 GH¢in 1 month

Source: [Transfer Project \(2020\)](#)

Table A3: Summary of empirical studies assessing the effects of LEAP 1000 in Ghana

Paper	Outcome	Method	Summary Findings
Palermo et al. (2019)	Enrolment in health insurance	Difference in differences, intention to treat effects; longitudinal qualitative analysis	Health insurance enrolment increased by 14% for adults and 15 % for children.
Otieno et al. (2022)	Effect of service availability on health insurance enrolment	Difference in differences, triple differences	Households in LEAP 1000 with higher levels of service availability had a 15% increase in the probability of enrolling in health insurance
de Groot et al. (2022)	Child nutrition (height for age and weight for age z-scores)	Difference in differences, intention to treat effects	No significant treatment effect on nutritional outcomes. Increase in household food security but reductions in child meal frequency
Quinones, Mendola, et al. (2023)	Average birth weight, low birth weight	Difference in differences	Low birth weight reduced by 3.5%, average birth weight increased by 94 grams overall
Quinones, Lin, et al. (2023)	Birth weight	Structural equation modelling	1 month increase in the exposure to LEAP was associated with increasing birth weight by 9 grams and a 7% reduction in probability of low birth weight. Mediating effect of household food security and mothers' agency was not statistically significant
Peterman et al. (2022)	Intimate partner violence	Difference in differences	No effects on the probability of experiencing intimate partner violence, reduction in the frequency of experiencing emotional or physical violence
de Milliano et al. (2021)	Social support	Difference in differences	increased social support, both emotional and instrumental social support. Increased participation in social support groups
LaPointe et al. (2024)	Low birth weight	Difference in differences	Among control households, increased weekly temperatures were associated with higher odds of low birth weight. Cash transfer mitigated the effect of temperature on low birth weight
Maara et al. (2023)	Self-perceived stress	Difference in differences	No significant effect of LEAP 1000 on self perceived stress
Novignon et al. (2022)	Sickness and health seeking	Difference in differences	Increased the probability of seeking care by 11 % moderate reduction in health expenditures

Table A4: Summary and baseline balance measures of control variables

Variable	(1) Comparison		(2) Treatment		(1)-(2) Pairwise t-test	
	N	Mean/(SE)	N	Mean/(SE)	N	P-value
Household size (number)	1146	6.347 (0.074)	1185	6.959 (0.078)	2331	0.125
Age of household head	1146	38.142 (0.362)	1185	40.451 (0.359)	2331	0.655
Eligible woman age	1146	28.614 (0.198)	1185	30.302 (0.196)	2331	0.600
Married Polygamous	1146	0.293 (0.013)	1185	0.305 (0.013)	2331	0.825
Married Monogamous	1146	0.647 (0.014)	1185	0.641 (0.014)	2331	0.586
Father Primary Education	1146	0.217 (0.012)	1185	0.183 (0.011)	2331	0.630
Mother Primary Education	1146	0.244 (0.013)	1185	0.245 (0.012)	2331	0.335
Household has electricity	1146	0.312 (0.014)	1185	0.263 (0.013)	2331	0.878
Poor quality housing	1146	0.532 (0.015)	1185	0.513 (0.015)	2331	0.528
Household has safe water	1146	0.576 (0.015)	1185	0.604 (0.014)	2331	0.826
Interpreter used	1146	0.087 (0.008)	1185	0.109 (0.009)	2331	0.946
F-test of joint significance (F-stat)						0.803
F-test, number of observations						2331

Pairwise and F-test regressions control for the proxy means test. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A5: Descriptive results of outcomes at baseline

Variable	(1) Comparison		(2) Treatment		(1)-(2) Pairwise t-test	
	N	Mean/(SE)	N	Mean/(SE)	N	P-value
Livestock ownership (1/0)	1235	0.508 (0.014)	1262	0.525 (0.014)	2497	0.736
Total livestock units (TLU)	1235	1.541 (0.227)	1262	1.472 (0.176)	2497	0.379
Large ruminants	1235	0.137 (0.010)	1262	0.135 (0.010)	2497	0.055*
Small ruminants	1235	0.361 (0.014)	1262	0.380 (0.014)	2497	0.801
Durable investments	1235	564.346 (16.725)	1262	570.618 (17.241)	2497	0.912
Business	1235	0.221 (0.012)	1262	0.204 (0.011)	2497	0.753
Savings	1235	0.090 (0.008)	1262	0.067 (0.007)	2497	0.016**
Loan	1235	0.464 (0.014)	1262	0.500 (0.014)	2497	0.190
Wage job	1235	0.268 (0.013)	1262	0.290 (0.013)	2497	0.468
Impatience (zscore)	1235	-0.144 (0.027)	1262	-0.071 (0.027)	2497	0.939
Future orientation(zscore)	1235	0.126 (0.027)	1262	0.068 (0.027)	2497	0.572
Always present	1235	0.259 (0.012)	1262	0.289 (0.013)	2497	0.544
Happiness (zscore)	1235	-0.083 (0.030)	1262	-0.166 (0.032)	2497	0.243
Expectation (zscore)	1235	-0.199 (0.034)	1262	-0.187 (0.034)	2497	0.777
Agency index (zscore)	1235	-0.327 (0.027)	1262	-0.294 (0.027)	2497	0.617
F-test of joint significance (F-stat)					1.344	
F-test, number of observations					2497	

Pairwise and F-test regressions control for the proxy means test. Significance: ***=.01, **=.05, *=.1

Table A6: Baseline balance of control variables between the treatments and control

Variable	(1) Control		(2) Cash + Insurance		(3) Cash only		(1)-(2)		(1)-(3) Pairwise t-test		(2)-(3)	
	N	Mean/(SE)	N	Mean/(SE)	N	Mean/(SE)	N	P-value	N	P-value	N	P-value
Household size (number)	2292	6.572 (0.057)	1742	6.613 (0.127)	624	6.552 (0.215)	4034	0.175	2916	0.112	2366	0.502
Age of household head	2292	40.458 (0.299)	1742	41.153 (0.717)	624	41.902 (1.257)	4034	0.155	2916	0.155	2366	0.443
Married Polygamous	2292	0.318 (0.010)	1742	0.307 (0.020)	624	0.331 (0.035)	4034	0.560	2916	0.676	2366	0.848
Married Monogamous	2292	0.618 (0.010)	1742	0.641 (0.022)	624	0.629 (0.038)	4034	0.610	2916	0.452	2366	0.800
Household has electricity	2292	0.348 (0.010)	1742	0.305 (0.019)	624	0.295 (0.033)	4034	0.810	2916	0.091*	2366	0.080*
Poor quality housing	2292	0.535 (0.011)	1742	0.564 (0.025)	624	0.586 (0.041)	4034	0.973	2916	0.265	2366	0.130
Father Primary Education	2292	0.201 (0.008)	1742	0.230 (0.035)	624	0.203 (0.062)	4034	0.525	2916	0.985	2366	0.824
Mother Primary Education	2292	0.261 (0.009)	1742	0.284 (0.034)	624	0.250 (0.059)	4034	0.038**	2916	0.329	2366	0.792
Interpreter used?	2292	0.066 (0.005)	1742	0.086 (0.010)	624	0.083 (0.014)	4034	0.752	2916	0.310	2366	0.625
F-test of joint significance (F-stat)								1.084		1.289		0.663
F-test, number of observations								4034		2916		2366

Pairwise and F-test regressions control for the proxy means test. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A7: Heterogenous effects of LEAP 1000 on livestock investments

VARIABLES	(1) Cattle	(2) Goats	(3) Sheep	(4) Poultry	(5) Large ruminants	(6) Small ruminants
Treatment	0.011 (0.014)	0.082*** (0.021)	0.008 (0.021)	0.091*** (0.024)	-0.003 (0.015)	0.099*** (0.022)
Constant	-0.059 (0.077)	-0.060 (0.107)	-0.070 (0.112)	0.207 (0.130)	-0.001 (0.078)	-0.065 (0.114)
Observations	4,658	4,658	4,658	4,658	4,658	4,658
R-squared	0.669	0.617	0.635	0.612	0.678	0.625

Pairwise regressions control for the Proxy Means Test, language of interview and community fixed effects. Significance levels correspond with *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A8: LATE heterogenous effects of LEAP 1000 on livestock investments

VARIABLES	(1) Cattle	(2) Goats	(3) Sheep	(4) Poultry	(5) Large ruminants	(6) Small ruminants
Cash + Insurance	0.013 (0.016)	0.099*** (0.027)	0.033 (0.024)	0.083** (0.033)	0.008 (0.017)	0.132*** (0.027)
Cash only	-0.016 (0.018)	0.017 (0.028)	-0.029 (0.026)	-0.064 (0.090)	-0.032* (0.019)	0.032 (0.030)
Constant	0.006 (0.094)	-0.047 (0.111)	-0.013 (0.136)	0.223 (0.138)	0.073 (0.086)	-0.033 (0.117)
Observations	4,658	4,658	4,658	4,658	4,658	4,658
R-squared	0.674	0.633	0.647	0.614	0.684	0.644

Pairwise regressions control for the Proxy Means Test, language of interview and community fixed effects. Significance levels correspond with *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Figure A1: Differences in baseline and endline prices for selected goods

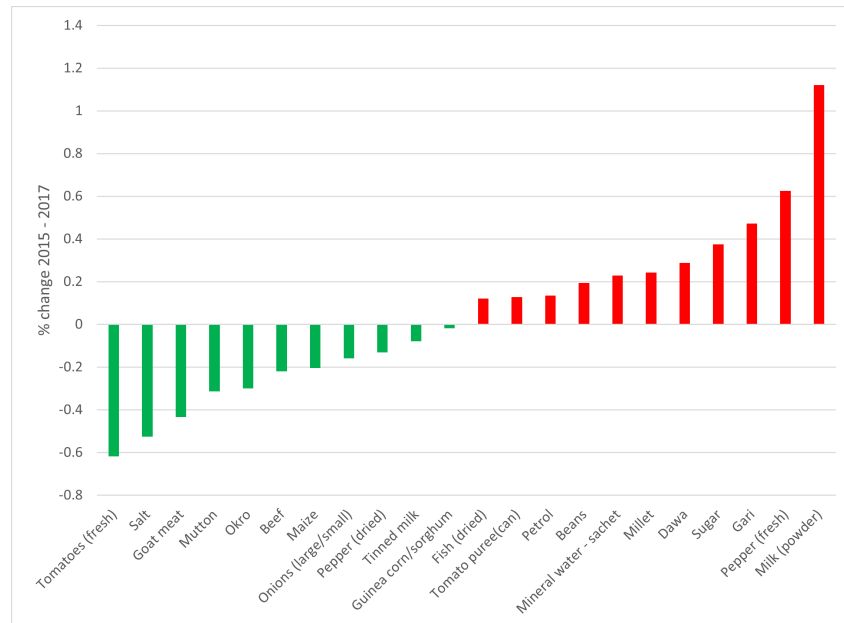


Figure A2: Distribution of Proxy Means Test scores by treatment group

